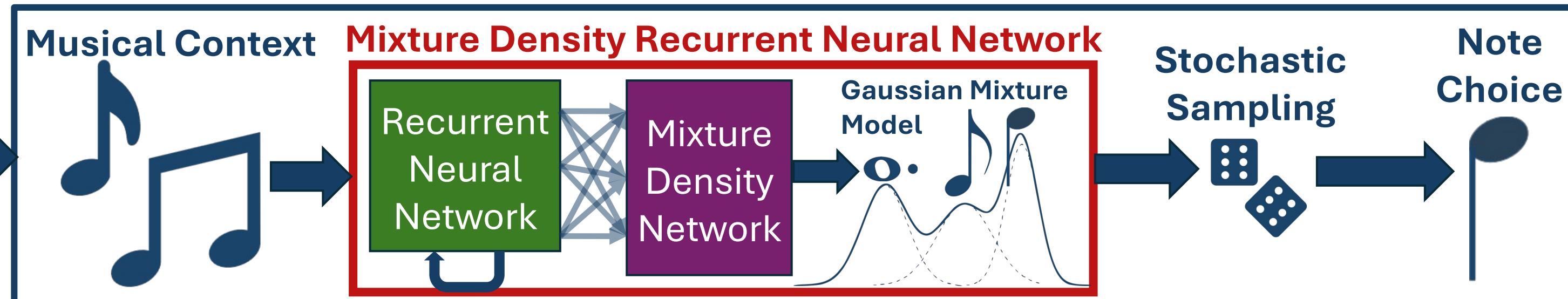


Beyond Stochastic Selection: Guiding Musical Predictions with Monte Carlo Tree Search

Scott Howsam, Charles Martin

Prior Prediction System

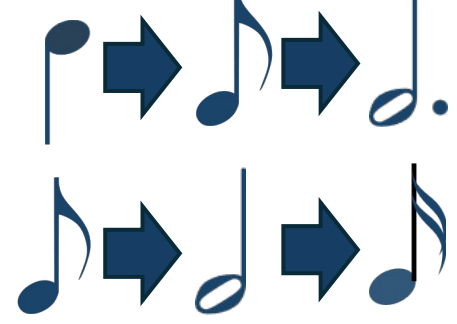
- A neural network provides multiple predictions of valid notes to play next, with probabilities for each
- Output is selected stochastically
- Surely there's a better way!



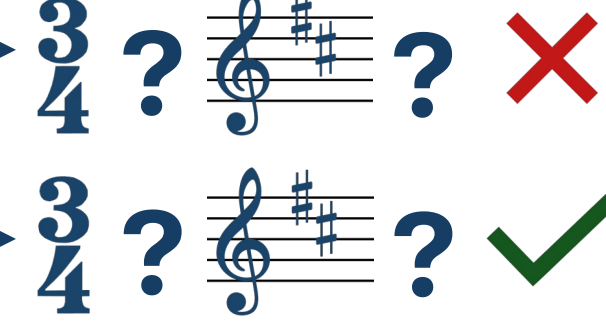
Random selection is arbitrary



We instead search future options



Assessing adherence to the broader musical structure

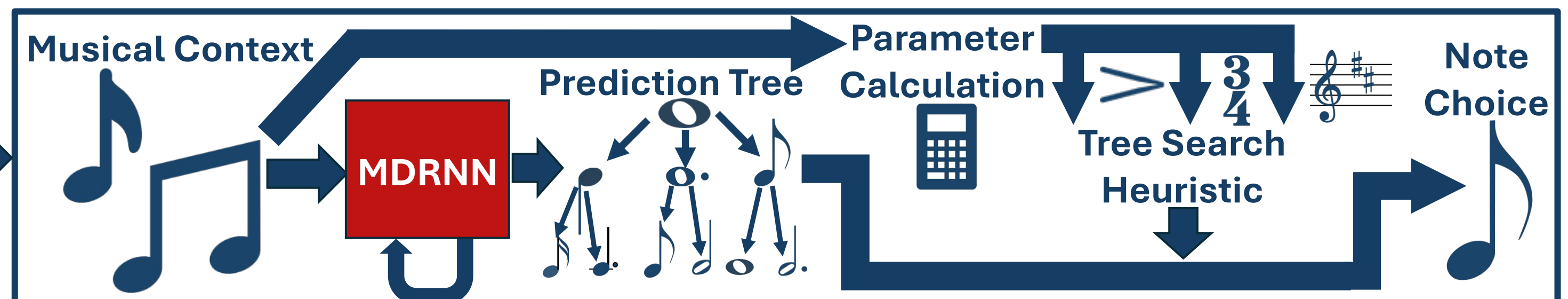


And hence generate better predictions



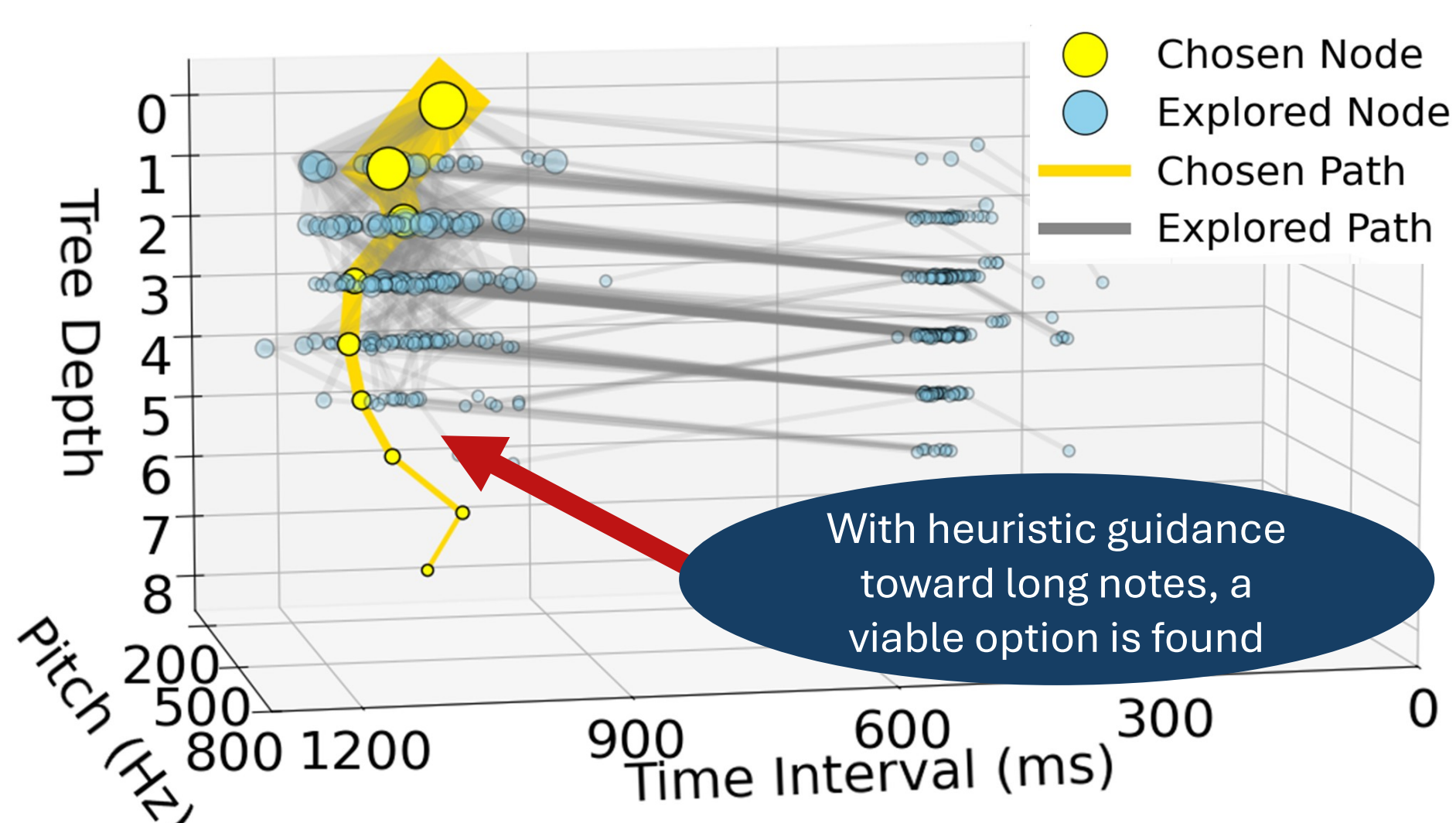
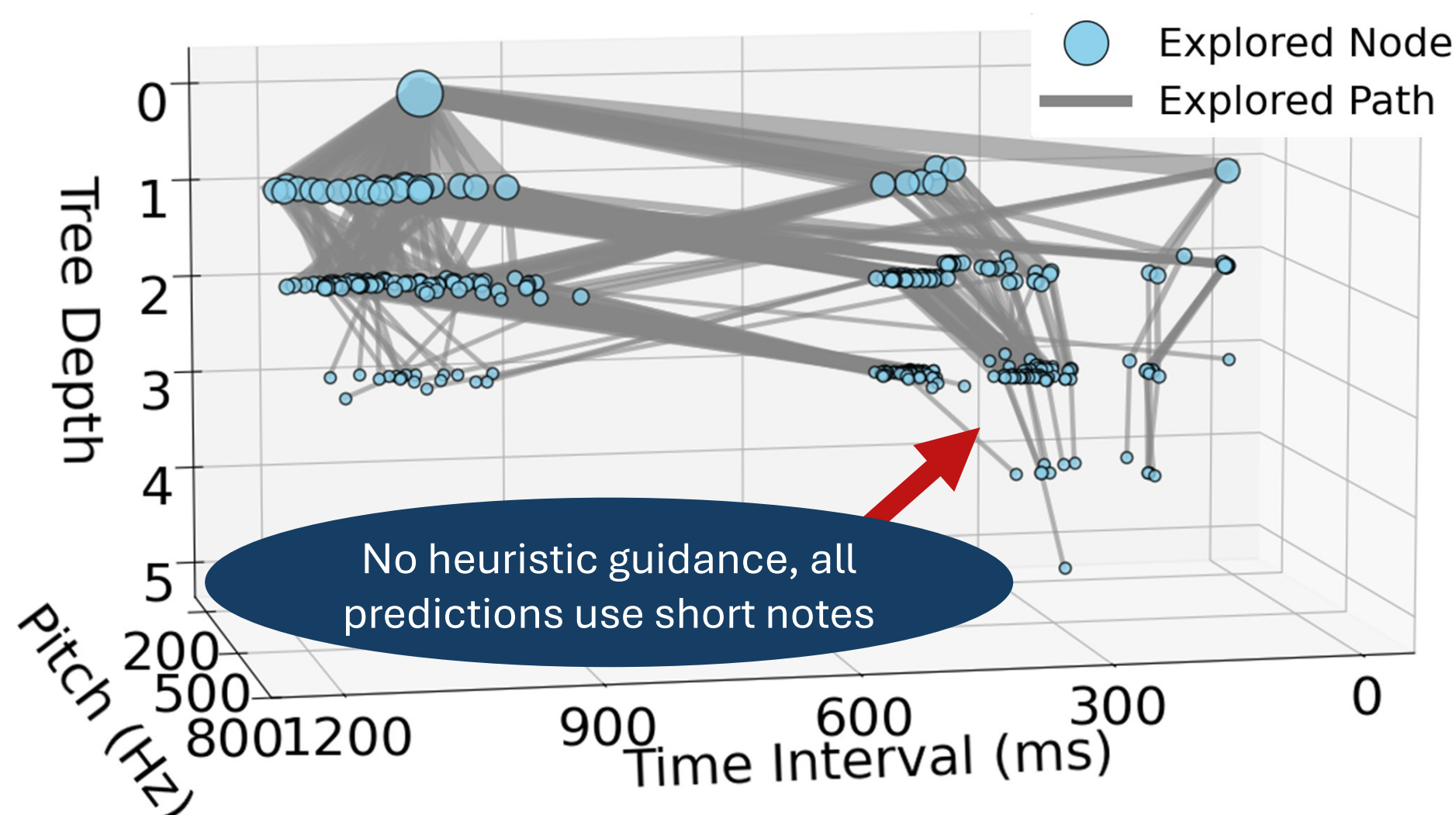
Our Prediction System

- Our method is closer to humans
- MDRNN generates multiple options
- Important parameters of the piece are generated from previous context
- Search for the best match to these



System Design

- Music has multiple valid continuations, not one 'correct' next note
- Therefore, AI musical prediction systems output a range of valid options
- To avoid stochastic selection, we create a Monte Carlo Search Tree of viable future musical sequences, and use a heuristic to rank them
- This heuristic is based on 'rules' humans implicitly use for prediction, such as tempo, key, and rhythm
- The first figure below shows a tree of musical paths with no guidance
- The second shows the same tree guided by a heuristic aiming to maintain long time intervals
- This heuristic is arbitrary, but displays the power these heuristics can have on the search process



Results

- Firstly, we evaluated the accuracy of our prediction system against a state-of-the-art published system on 11,142 unseen notes across 15 improvisations. While not every parameter improved performance for every improvisation, on average the heuristic improved accuracy by over 30%.
- We also wished to maintain the creativity, originality, and human-like nature of the outputs. For this, a qualitative analysis of multiple improvisation sessions with the AI (an example of which is shown in the figure below) revealed that our method did not sacrifice creativity. It provided a better collaborative improvisational experience than previous methods.



Conclusion

- Integrating Monte Carlo Tree Search into musical prediction systems improves their performance significantly in multiple metrics
- Multiple future work avenues seem very promising, including a UI for further user control, and many alternative heuristics